INCH-POUND

MIL-DTL-55302/180C
w/AMENDMENT 1
15 April 2013
SUPERSEDING
MIL-DTL-55302/180C
23 January 2008

DETAIL SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: 4 ROW, RECEPTACLE ASSEMBLIES, RIGHT ANGLE, 100 THROUGH 684 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

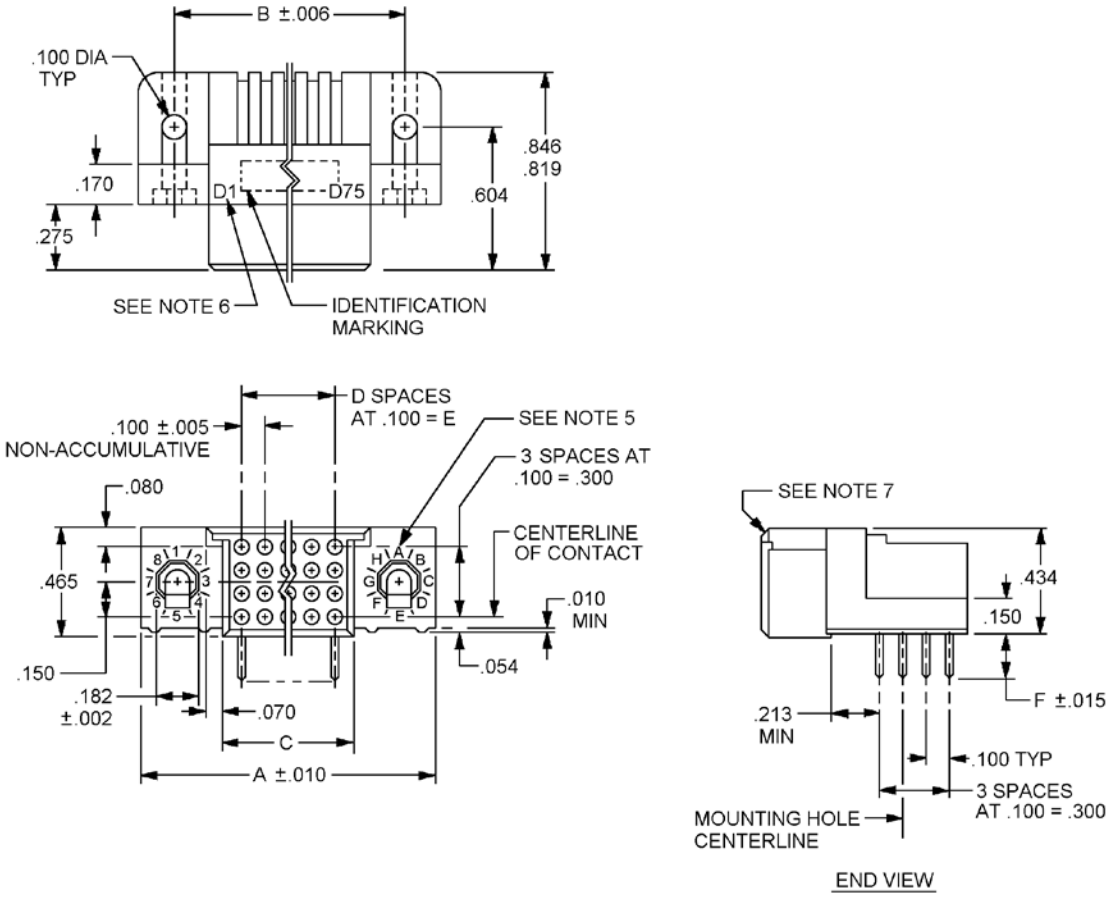
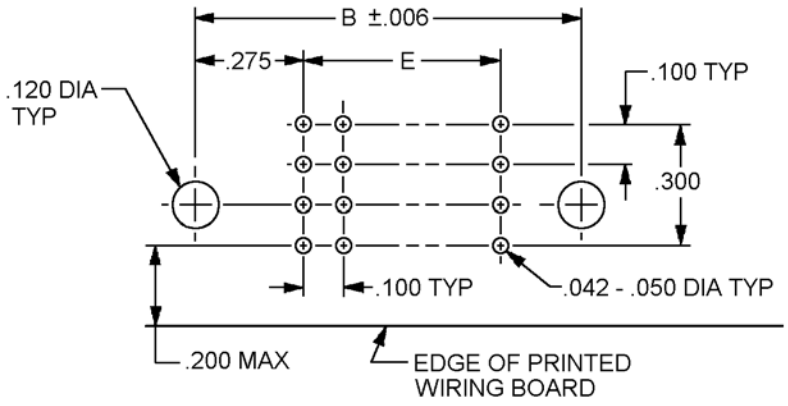


FIGURE 1. Connectors, receptacle assemblies .100 (2.54 mm) spacing) 100 through 300 contacts.

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F ± .015 (0.38 mm)		Type	Type of termination
Inches	mm		
.180	4.57	I	Solder post (see note 8)
.120	3.05	IV	

Inches	mm	Inches	mm
.002	0.05	.150	3.81
.005	0.13	.170	4.32
.006	0.15	.182	4.62
.010	0.25	.200	5.08
.015	0.38	.213	5.41
.042	1.07	.275	6.98
.050	1.27	.300	7.62
.054	1.37	.434	11.02
.070	1.78	.465	11.81
.080	2.03	.604	15.34
.100	2.54	.819	20.80
.120	3.05	.846	21.49

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm) on decimals and ± 2° on angles.
4. These connectors mate with connectors specified in MIL-DTL-55302/179 and /181.
5. Location indicators embossed on surface.
6. Numbers indicating end cavities, letters indicating row nearest to polarizing feature, and markings every ten positions stamped on this surface.
7. Chamfers shall be .015 (0.38 mm) minimum at a 45° angle ± 5°.
8. Solder post shall be a .036 (0.91 mm) maximum diameter or .036 (0.91 mm) maximum across the diagonal.

FIGURE 1. Connectors, receptacle assemblies .100 (2.54 mm) spacing 100 through 300 contacts – Continued.

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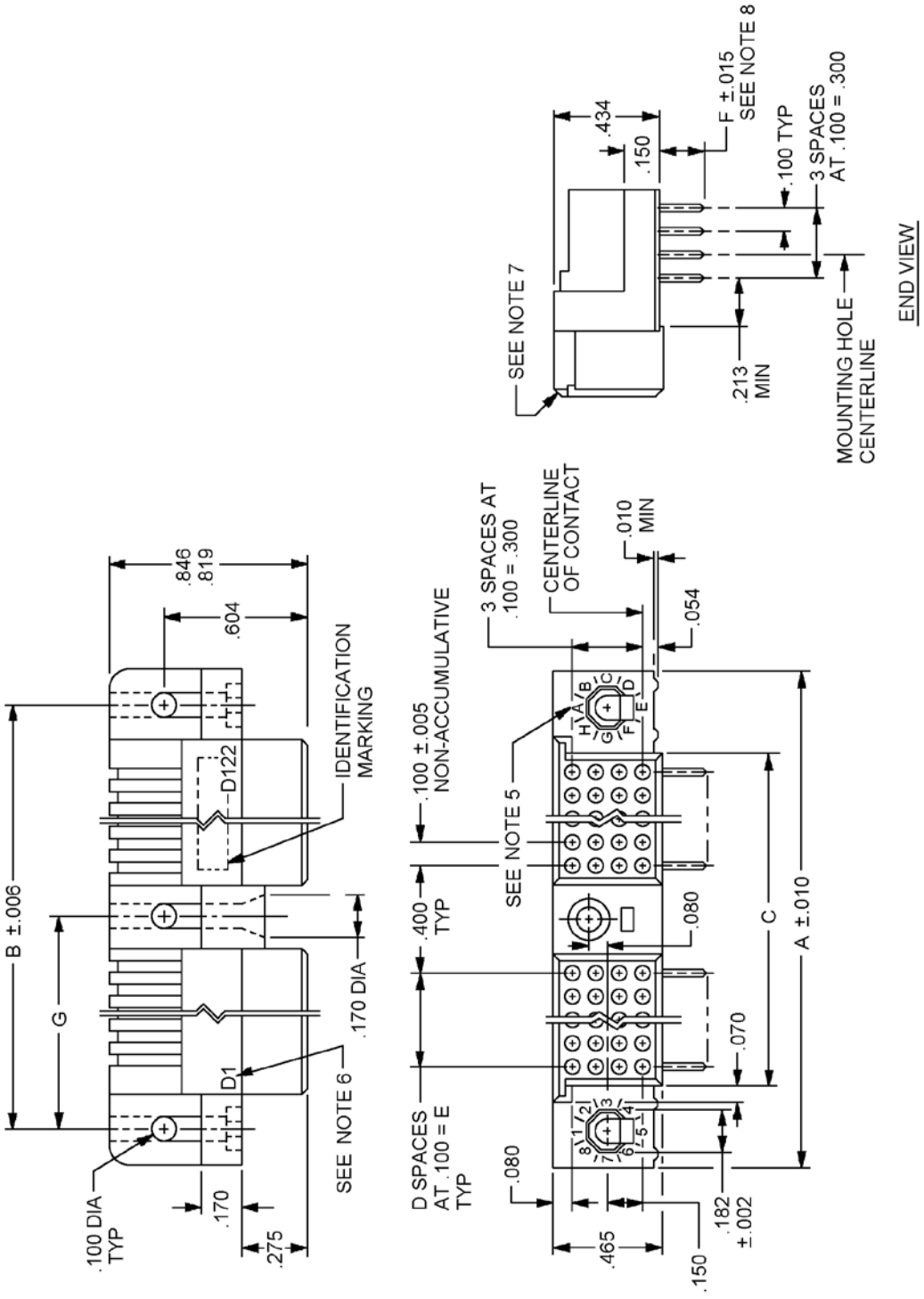
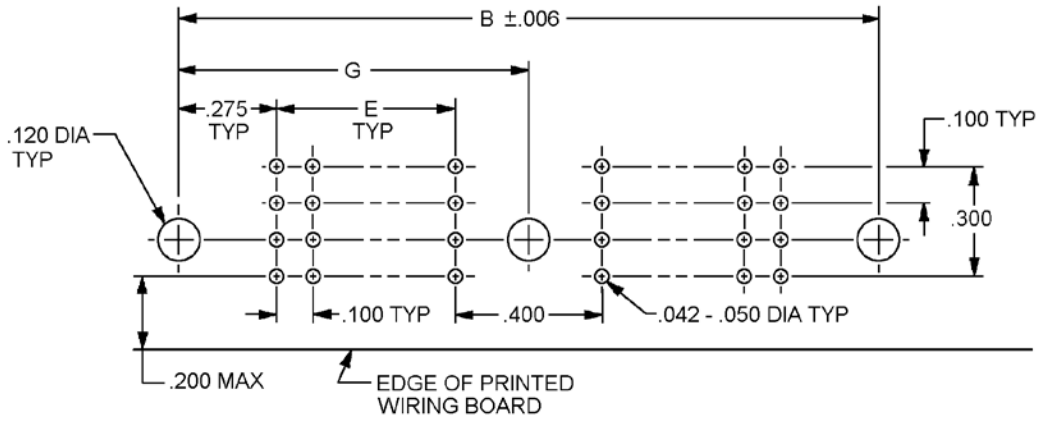


FIGURE 2. Connectors, receptacle assemblies .100 (2.54 mm) spacing with one guide, 320 through 488 contacts.

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F ± .015 (0.38 mm)		Type	Type of termination
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5. Location indicators embossed on surface.
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7. Chamfers shall be .015 (0.38 mm) minimum at a 45° angle ± 5°.
8. Solder post shall be a .036 (0.91 mm) maximum diameter or .036 (0.91 mm) maximum across the diagonal.

FIGURE 2. Connectors, receptacle assemblies .100 (2.54 mm) spacing with one guide, 320 through 488 contacts – Continued.

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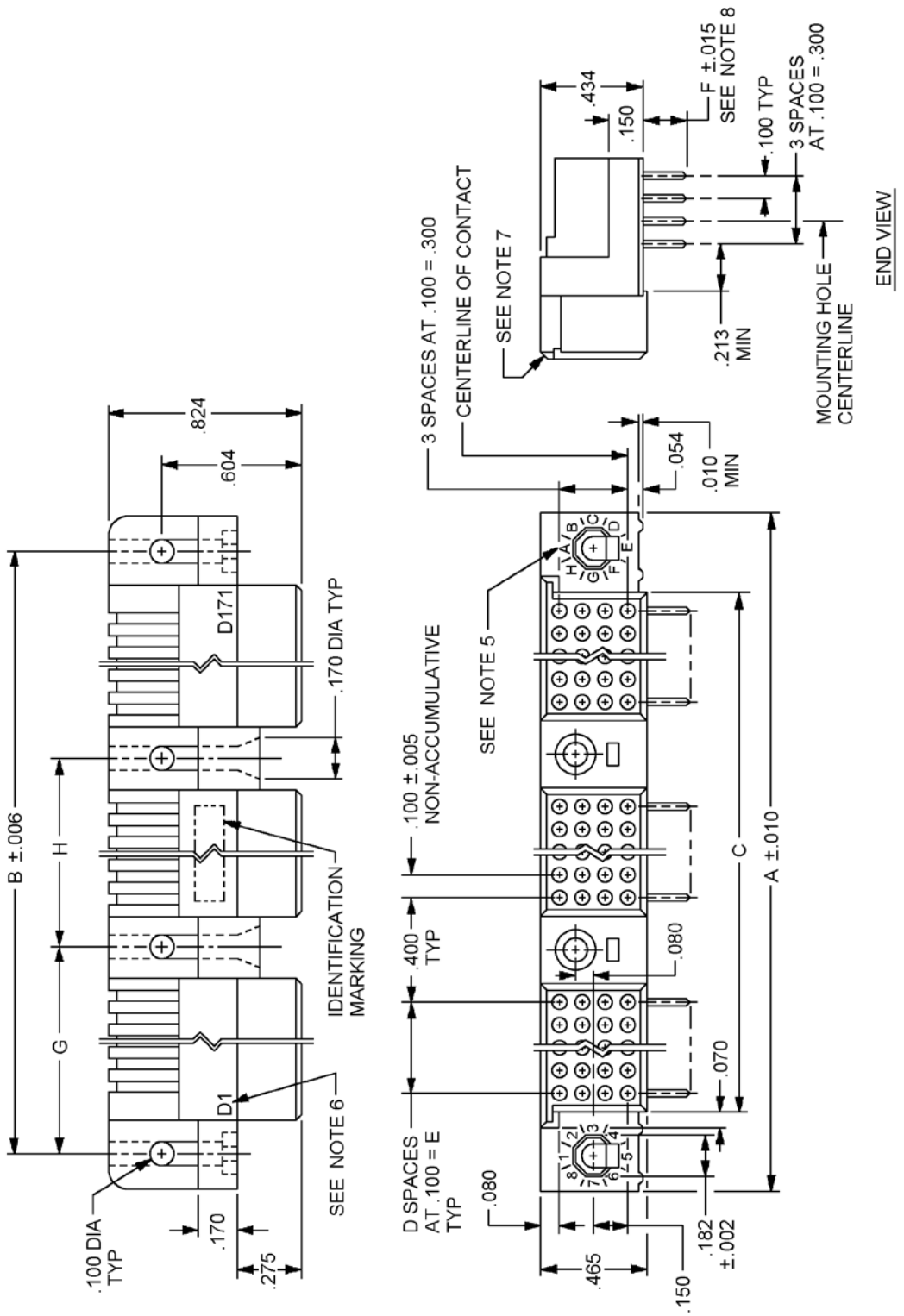
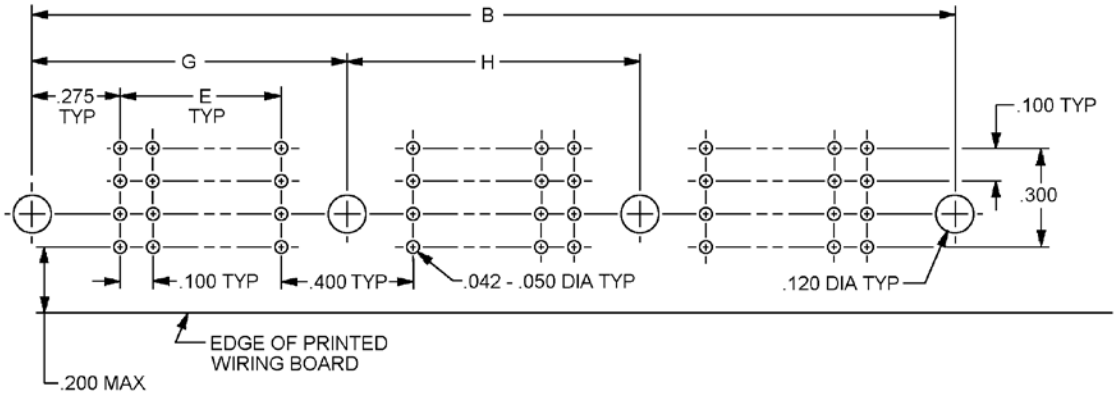


FIGURE 3. Connectors, receptacle assemblies .100 (2.54 mm) spacing with two guide, 516 through 684 contacts.

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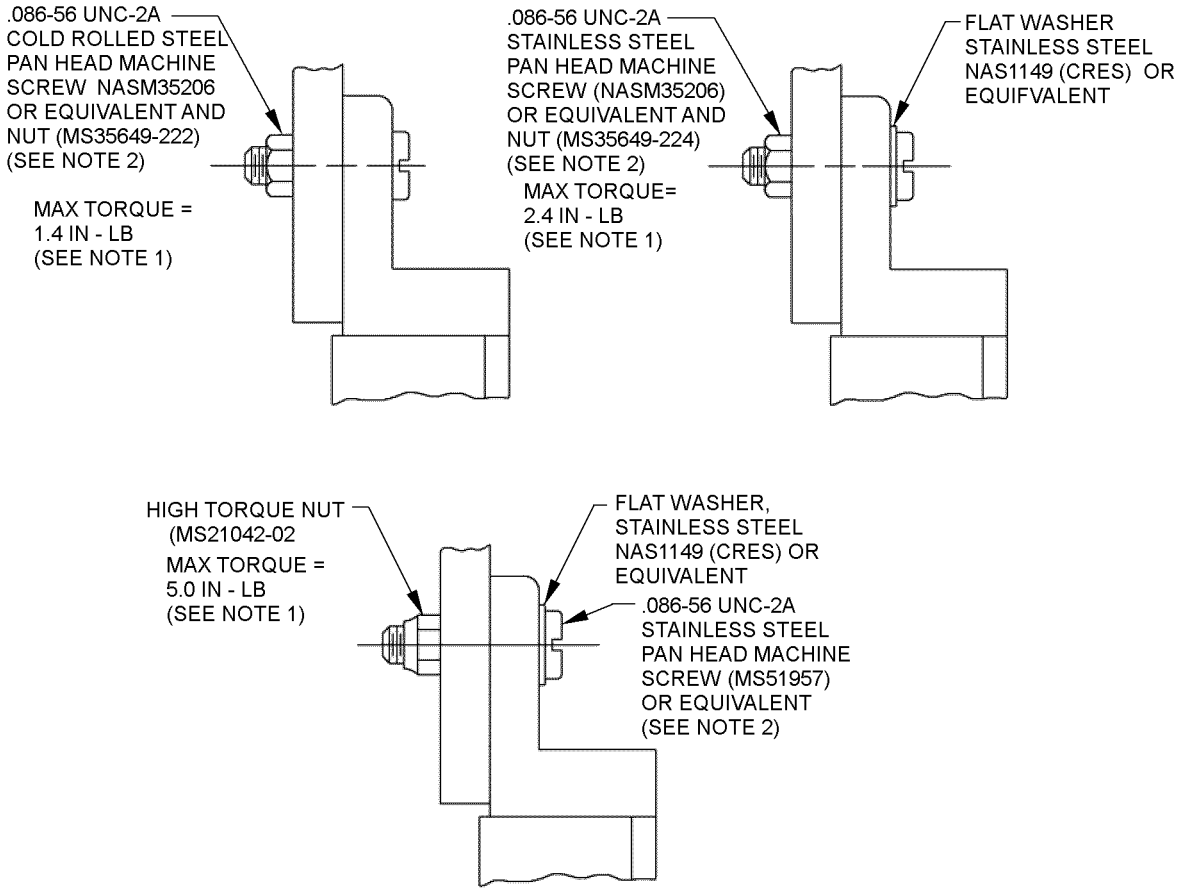
F ± .015 (0.38 mm)		Type	Type of termination
Inches	mm		
.180	4.57	I	Solder post (see note 8)
.120	3.05	IV	

Inches	mm	Inches	mm
.002	0.05	.150	3.81
.005	0.13	.170	4.32
.006	0.15	.182	4.62
.010	0.25	.200	5.08
.015	0.38	.213	5.41
.042	1.07	.275	6.98
.050	1.27	.300	7.62
.054	1.37	.400	10.16
.070	1.78	.434	11.02
.080	2.03	.465	11.81
.100	2.54	.604	15.34
.120	3.05	.824	20.93

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm) on decimals and ± 2° on angles.
4. These connectors mate with connectors specified in MIL-DTL-55302/179 and /181.
5. Location indicators embossed on surface.
6. Numbers indicating end cavities, letters indicating row nearest to polarizing feature, and markings every ten positions stamped on this surface.
7. Chamfers shall be .015 (0.38 mm) minimum at a 45° angle ± 5°.
8. See figure 1 for solder post types. Solder post shall be a .036 (0.91 mm) maximum diameter or .036 (0.91 mm) maximum across the diagonal.

FIGURE 3. Connectors, receptacle assemblies .100 (2.54 mm) spacing with two guides, 516 through 684 contacts – Continued.

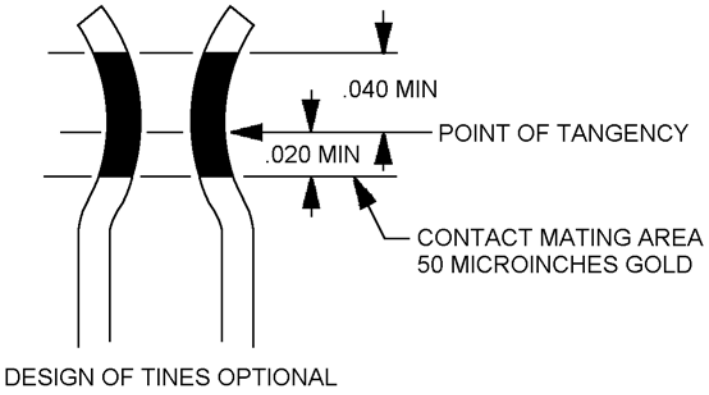
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NOTES:
1. Torque wrench must be used when fastening connector to printed circuit board.
2. Length to be determined by user.

FIGURE 4. Mounting recommendations.

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Inches	mm
.020	.051
.040	1.02

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.

FIGURE 5. Contact plating.

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TABLE I. Dash number, number of contacts, contact type, and dimensions. 1/ 2/

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
01 04	100	1	I IV	3.250 (82.55)	2.950 (74.93)	2.524 (64.11)	24	2.400 (60.96)			1-25
09 12	128	1	I IV	3.950 (100.33)	3.650 (92.71)	3.224 (81.89)	31	3.100 (78.74)			1-32
17 20	160	1	I IV	4.750 (120.65)	4.450 (113.03)	4.024 (102.21)	39	3.900 (99.06)			1-40
25 28	200	1	I IV	5.750 (146.05)	5.450 (138.43)	5.024 (127.61)	49	4.900 (124.46)			1-50
29 32	220	1	I IV	6.250 (158.75)	5.950 (151.13)	5.524 (140.31)	54	5.400 (137.16)			1-55
33 36	240	1	I IV	6.750 (171.45)	6.450 (163.83)	6.024 (153.01)	59	5.900 (149.86)			1-60
37 40	260	1	I IV	7.250 (184.15)	6.950 (176.53)	6.524 (165.71)	64	6.400 (162.56)			1-65
41 44	268	1	I IV	7.450 (189.23)	7.150 (181.61)	6.724 (170.79)	66	6.600 (167.64)			1-67
45 48	280	1	I IV	7.750 (196.85)	7.450 (189.23)	7.024 (178.41)	69	6.900 (175.26)			1-70
49 52	300	1	I IV	8.250 (209.55)	7.950 (201.93)	7.524 (191.11)	74	7.400 (187.96)			1-75
57 60	344	2	I IV	9.650 (245.11)	9.350 (237.49)	8.924 (226.67)	42	4.200 (106.68)	4.675 (118.75)		1-86
69 72	416	2	I IV	11.450 (290.83)	11.150 (283.21)	10.724 (272.39)	51	5.100 (129.54)	5.575 (141.61)		1-104
81 84	488	2	I IV	13.250 (336.55)	12.950 (328.93)	12.524 (318.11)	60	6.000 (152.40)	6.475 (164.46)		1-222

See footnote at the end of table.

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TABLE I. Dash number, number of contacts, contact type, and dimensions - Continued. 1/ 2/

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
85 88	516	3	I IV	14.250 (361.95)	13.950 (354.33)	13.524 (343.51)	42	4.200 (106.68)	4.675 (118.75)	4.600 (116.84)	1-129
89 92	540	3	I IV	14.850 (377.19)	14.550 (369.57)	14.124 (358.75)	44	4.400 (111.76)	4.875 (123.83)	4.800 (121.92)	1-135
101 104	612	3	I IV	16.650 (422.91)	16.350 (415.29)	15.924 (404.47)	50	5.000 (127.00)	5.475 (139.07)	5.400 (137.16)	1-153
105 108	636	3	I IV	17.250 (438.15)	16.950 (430.53)	16.524 (419.71)	52	5.200 (132.08)	5.675 (144.15)	5.600 (142.24)	1.159
113 116	684	3	I IV	18.450 (468.63)	18.150 (461.01)	17.724 (450.19)	56	5.600 (142.24)	6.075 (154.31)	6.000 (152.40)	1-171

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.

TABLE II. Contact plating requirements. 1/

Type	Finish requirement
A	Overall contact finish: Shall be gold in accordance with MIL-DTL-45204, class 1, type II, grade C or equivalent.
B	Localized contact finish: Contact engagement area shall have a minimum plating (see figure 4) in accordance with MIL-DTL-45204, class 1, type II, grade C or equivalent. Solder post area (contact type I, .195 minimum; length; type IV, .135 minimum length) shall be tin-lead over nickel in accordance with MIL-DTL-55302.

1/ Type C finish is deleted and superseded by type A or B.

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REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figures 1, 2, 3, 4 and table I.

Materials: Shall be in accordance with MIL-DTL-55302.

Contacts: Shall be copper-beryllium in accordance with ASTM B194, ASTM B196/B196M, or ASTM B197/B197M.

Plating: See table II and figure 5.

Current rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at room ambient with no more than two adjacent contacts carrying this current.

Keying (see MIL-DTL-55302/31):

Two keys, Part or Identifying Number (PIN) M55302/31-04, and two retaining rivets, PIN M55302/31-10 or two .086-56 UNC-2A mounting screws, are recommended. Keys and rivets or screws shall be ordered separately.

Jackscrews (see MIL-DTL-55302/182):

Use M55302/182-07, or –09. Jackscrews are purchased separately.

Mating and unmating: The mating force in pounds shall be the number of contacts multiplied by .250; the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: No individual contact pair shall have a resistance exceeding .015-ohm initial and .020 after testing.

Contact retention: Not applicable.

Dielectric withstanding voltage:

Sea level: 900 volts rms.

70,000 feet: 200 volts rms.

2 milliamperes maximum leakage current.

PIN: M55302/180 (and dash number from table I and type from table II).

Group qualification: See table III.

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TABLE III. Group qualification.

Qualification of any of the following connectors 1/ 2/	Qualifies the following connectors	Retains qualification on the following
M55302/174-***	M55302/174-***	M55302/174-*** M55302/177-*** M55302/180-***
M55302/177-***	M55302/174-*** M55302/177-***	
M55302/180-***	M55302/174-*** M55302/177-*** M55302/180-***	

1/ For initial qualification, connectors with the largest number of contact positions from the left hand column shall be used to obtain qualification for parts in corresponding center column.
2/ For qualification retention data may be supplied on any two parts from the left hand column in order to retain qualification for the parts listed in the right hand column.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/31
MIL-DTL-55302/179
MIL-DTL-55302/181
MIL-DTL-55302/182
MIL-DTL-45204
ASTM B194
ASTM B196/B196M
ASTM B197/B197M
MS51957
NAS1149
NASM35206

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2013-068)

Review activities:
Army - AR, AT, AV, CR4, MI
Navy - AS, MC, SH
Air Force - 19, 99

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